



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY**

**ACADEMIC QUALITY ASSURANCE
HANDBOOK**

FOREWORD

Jomo Kenyatta University of Agriculture and Technology (JKUAT) has enshrined quality as one of its beacons in the development and implementation of academic programs. This is done with close reference to the guidelines issued by the Commission for University Education (CUE), which is the rational body charged with responsibility of standardization of University Education in Kenya.

JKUAT has a quality Assurance Policy whose implementation is overseen by the Directorate of Academic Quality Assurance. The Policy encapsulates all the activities from curriculum development through issuance of results to candidates in various programmes. Mechanisms of correcting any deviations to the maintenance and development of quality are also outlined.

Through DAQA, JKUAT has developed an Academic Quality Assurance Handbook which is a valuable solution to the literature available on the management of academic quality assurance. It has significantly defined terms, sequenced activities and developed guidelines for monitoring and evaluation of quality related activities. This is a commendable achievement considering that most of the staff implementing the curriculum have moderate pedagogical exposure. For those with adequate pedagogical skills, the handbook will serve to refresh their thinking on skills outlined and probably contribute to future editions of the handbook.

I wish to recognize the guidance that DAQA received from both deans and senate in the development of the handbook. It is my sincere hope that this becomes a well-read book that will steer the university steadily on its chosen path of quality.

I take this opportunity to flag off this first Edition of the Academic Quality Assurance Handbook in anticipation that will serve as a baseline for continuous academic quality improvement.

Prof. Victoria W. Ngumi

Vice Chancellor

PREFACE

The University is the cradle of civilization. This is the space where knowledge is created and disseminated. This implies then that proper infrastructure, both physical and human, must be organized to ensure that the available knowledge is not drained to waste. The directorate of Academic quality Assurance in Jomo Kenyatta University of Agriculture and Technology (JKUAT) was created to ensure that the ‘prescribed’ knowledge is in accordance with the stakeholders’ needs and solution based to market needs assessment.

This handbook is a guideline of what is expected to be done to ensure that JKUAT delivers quality education in its area of mandate. This will guarantee that all the certificates, diplomas and degrees awarded will be of the highest educational standards. The graduates released into the world will be knowledgeable, creative and innovative.

I thank all the Members of the Quality Assurance Committee who spent tireless moments to critique the contents of this handbook.

Prof. Okong’o Eric Rang’ondi,
Director, Directorate of Academic Quality Assurance

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INTRODUCTION

It is impossible to explain what academic quality assurance is without knowing what is quality. Webster Dictionary defines quality in different aspects. Among these is Nature: peculiar and essential character; Property: an inherent feature; Grade: degree of excellence; Characteristic: a distinguishing attribute etc. Everybody knows what quality is and can explain about quality. An absolute definition of quality does not exist because quality depends on the mind and expectation of whoever is defining it.

There are many interested parties in the Universities. These can be termed as stakeholders. Among these are:

- i. Students
- ii. Parents
- iii. Industries
- iv. Research organizations
- v. Financial institutions
- vi. Employers
- vii. The University Council
- viii. The community surrounding the University.

Each of these stakeholders has a different expectation on the University and will define quality in terms of the expectation. This makes quality a multifaceted aspect. In Mathematics, quality can be defined as a multivariate function whose output will depend on the inputs. Universities can be seen as standing at the crossroad. Every stakeholder wants the University to follow its direction of interest.

Role of the University in Defining Quality

Sometimes a patient goes to hospital and is diagnosed as suffering from several infections arising from different strains of bacteria. Each of these bacteria require a specific antibiotic. To treat the patient, the doctor will prescribe a broad spectrum antibiotic. As an analogy, Universities are faced with several interests arising from the different stakeholders. They

must prescribe a broad spectrum antibiotic which will ‘heal’ the different interests synchronously without biasness. This it does through development of Curriculum with objectives of meeting the needs of the stakeholders. It is a gross misconduct to develop a curriculum without identifying the stakeholders and addressing their needs. It will be like tailoring an outfit and forcing somebody to wear it. It might be unfit in size, colour, material, design, etc.

Quality is therefore a negotiation between the stakeholder and the University. The stakeholder must clearly identify her needs. The University through its faculty must formulate the Vision, Mission, objectives and goals of meeting the needs of the stakeholder through the educational programmes developed and the research activities. If this is achieved, then quality is archived in all its multi-facets.



Figure 1: Flow chart of information from stakeholder to implementation

Quality thus can be defined as achieving the goals and objectives effectively and efficiently assuming that the stakeholders' requirements are adequately addressed.

Role of the Quality Assurance Handbook

This handbook might not address all the requirements for Academic quality assurance. It's meant to arouse the interest in all University Academic staff to develop and embrace standards in monitoring and evaluation in our quality assurance tools. The areas that have been covered include:

- i. Effective teaching and learning
- ii. Setting and moderating examinations
- iii. Curriculum development and review
- iv. Monitoring and evaluation

CHAPTER ONE

1.0: Effective Teaching and Learning

In classroom teaching, it is necessary to teach using any approved teaching methods e.g. demonstrations, lecturing, presentations, practicals etc. To be meaningful, it is necessary to ensure that the intended objectives, here referred as the learning outcomes have been achieved. If the learning outcomes are achieved, then it can be concluded that Learning has taken place and teaching has been effective. Learning is an overall change of behavior. It can be demonstrated by the learner becoming knowledgeable, applying new skills, being creative, and ability to express self in the domain of specialization.

1.1: Bloom's Taxonomy of Learning Domains

Bloom's Taxonomy was created in 1956 under the leadership of educational psychologist Dr Benjamin Bloom in order to promote higher forms of thinking in education, such as analyzing and evaluating concepts, processes, procedures, and principles, rather than just remembering facts. It is used when designing educational, training, and learning processes.

1.2: Bloom's Learning Domains

Bloom's committee identified three *domains* of educational activities or learning (Bloom, et al. 1956):

1. **Cognitive:** mental skills (*knowledge*)
2. **Affective:** growth in feelings or emotional areas (*attitude or self*)
3. **Psychomotor:** manual or physical skills (*skills*)

This taxonomy of learning behaviors may be thought of as “the goals of the learning process.” That is, after a learning episode, the learner should have acquired a new skill, knowledge, and/or attitude.

1.2.1: The Cognitive Domain

The cognitive domain involves knowledge and the development of intellectual skills. This includes the recall or recognition of specific facts, procedural patterns, and concepts that serve in the development of intellectual abilities and skills. There are six major levels of cognitive domain processes, starting from the simplest to the most complex:

- i. Knowledge
- ii. Comprehension
- iii. Application
- iv. Analysis
- v. Synthesis
- vi. Evaluation

The levels can be thought of as building on one another. That is, the first ones must normally be mastered before the next one can take place.

1.2.2: Bloom's Cognitive Revised Taxonomy

Lorin Anderson, a former student of Bloom, and David Krathwohl revisited the cognitive domain in the mid-nineties and made some changes:

- i. changing the names in the six categories from noun to verb forms
- ii. rearranging them as shown in the chart below
- iii. creating a processes and levels of knowledge matrix

The revised Bloom's Revised Taxonomy has the following hierarchical categories:

- i. Creativity
- ii. Evaluation
- iii. Analysis
- iv. Application
- v. Understanding
- vi. Remembering

The new version of Bloom's Taxonomy, with examples and keywords is shown in the table below:

Table 2.1 Revised Cognitive domain

Category	Examples, key words (verbs), for learning (activities)
Remembering: Recall or retrieve previous learned information	Examples: Recite a policy. Quote prices from memory to a customer. Recite the safety rules. Key Words: defines, describes, identifies, knows, labels, lists, matches, names, outlines, recalls, recognizes, reproduces, selects, states
Understanding: Comprehending the meaning, translation, interpolation, and interpretation of instructions and problems. State a problem in one's own words	Examples: Rewrite the principles of test writing. Explain in one's own words the steps for performing a complex task. Translate an equation into a computer spreadsheet. Key Words: comprehends, converts, defends, distinguishes, estimates, explains, extends, generalizes, gives an example, infers, interprets, paraphrases, predicts, rewrites, summarizes, translates.
Applying: Use a concept in a new situation or unprompted use of an abstraction. Applies what was learned in the classroom into novel situations in the work place.	Examples: Use a manual to calculate an employee's vacation time. Apply laws of statistics to evaluate the reliability of a written test. Key Words: applies, changes, computes, constructs, demonstrates, discovers, manipulates, modifies, operates, predicts, prepares, produces, relates, shows, solves, uses.

Analyzing: Separates material or concepts into component parts so that its organizational structure may be understood. Distinguishes between facts and inferences.	Examples: Troubleshoot a piece of equipment by using logical deduction. Recognize logical fallacies in reasoning. Gathers information from a department and selects the required tasks for training. Key Words: analyzes, breaks down, compares, contrasts, diagrams, deconstructs, differentiates, discriminates, distinguishes, identifies, illustrates, infers, outlines, relates, selects, separates.
Evaluating: Make judgments about the value of ideas or materials.	Examples: Select the most effective solution. Hire the most qualified candidate. Explain and justify a new budget. Key Words: appraises, compares, concludes, contrasts, criticizes, critiques, defends, describes, discriminates, evaluates, explains, interprets, justifies, relates, summarizes, supports.
Creating: Builds a structure or pattern from diverse elements. Put parts together to form a whole, with emphasis on creating a new meaning or structure.	Examples: Write a company operations or process manual. Design a machine to perform a specific task. Integrates training from several sources to solve a problem. Revises and process to improve the outcome. Key Words: categorizes, combines, compiles, composes, creates, devises, designs, explains, generates, modifies, organizes, plans, rearranges, reconstructs, relates, reorganizes, revises, rewrites, summarizes, tells, writes

When Learners are introduced into a new field, they are expected to remember the fundamental terminologies in that field. For example, first

year student in Chemistry learning atomic structure is expected to remember the constituents of the atom, the Bohr structure of the atom and the quantum theory of the structure of the atom.

The student will proceed to bonding with an understanding why bonding takes place. He will then use the Schrodinger equation to generate acceptable wave functions defining orbitals. Further, the student will apply the linear combination of atomic orbitals (LCAO) to explain bonding and formation of Molecular orbitals. This is an illustration how the levels of the cognitive domain build on one another.

1.3: The Affective Domain

In teaching, the cognitive domain is emphasized. The affective domain can significantly enhance, inhibit or even prevent student learning. It includes factors such as student motivation, attitudes, perceptions and values. Teachers can increase their effectiveness by considering the affective domain in planning courses, delivering lectures and activities, and assessing student learning.

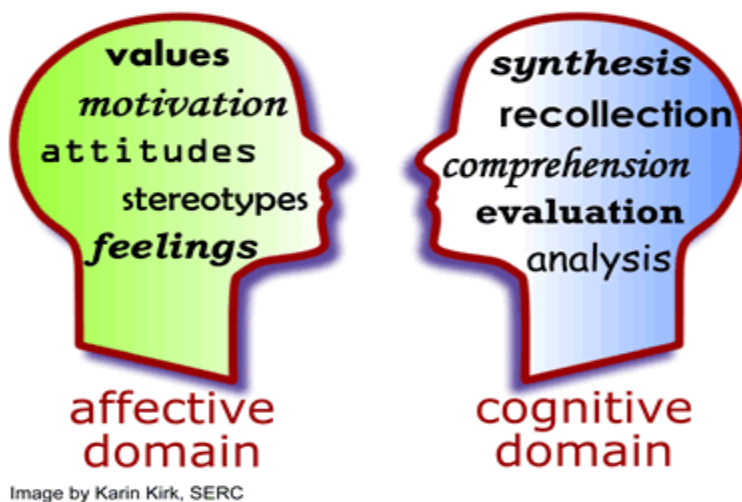


Figure 2: Facets of the affective and cognitive domains

The affective domain describes learning objectives that emphasize a feeling tone, an emotion, or a degree of acceptance or rejection. Affective objectives vary from simple attention to selected phenomena to complex but internally consistent qualities of character and conscience.

1.3.1: Krathwol's affective domain taxonomy

Krathwol's affective domain taxonomy is perhaps the best known of any of the affective taxonomies. "The taxonomy is ordered according to the principle of internalization. Internalization refers to the process whereby a person's affect toward an object passes from a general awareness level to a point where the affect is 'internalized' and consistently guides or controls the person's behavior (Seels & Glasgow, 1990, p. 28)."

1.3.2: Five stages of the Krathwol's Affective Domain

- i. **Receiving:** describes the stage of being aware of or sensitive to the existence of certain ideas, material, or phenomena and being willing to tolerate them. Examples include: to differentiate, to accept, to listen (for), and to respond to.
- ii. **Responding:** describes the second stage of the taxonomy and refers to a commitment in some small measure to the ideas, materials, or phenomena involved by actively responding to them. Examples are: to comply with, to follow, to commend, to volunteer, to spend leisure time in, to acclaim.
- iii. **Valuing:** means being willing to be perceived by others as valuing certain ideas, materials, or phenomena. Examples include: to increase measured proficiency in, to relinquish, to subsidize, to support, to debate.
- iv. **Organization:** involves relating the new value to those one already holds and bringing it into a harmonious and internally consistent philosophy. Examples are: to discuss, to theorize, to formulate, to balance, to examine.

- v. **Characterization** by value or value set means acting consistently in accordance with the values the individual has internalized. Examples include: to revise, to require, to be rated high in the value, to avoid, to resist, to manage, to resolve.

1.3.3: Relevance of the Affective domain to Learning

Teachers must be interested in students' attitudes toward learning new ideas, skills, philosophies and any other new information. They must find teaching methods that encourage Learners and draw them in. Affective topics in educational literature include attitudes, motivation, communication styles, classroom management styles, learning styles, use of technology in the classroom and nonverbal communication. It is also important not to turn students off by subtle actions or communications that go straight to the affective domain and prevent students from becoming engaged.

It is generally agreed that the affective domain is essential for learning, but it is the least studied, most often overlooked and the hardest to evaluate of Bloom's three domains. In formal classroom teaching, the teacher's efforts are typically focused on the cognitive aspects of the teaching and learning and most of the classroom time is designed for cognitive outcomes. It is also equally important to note that evaluating cognitive learning is straightforward but assessing affective outcomes is difficult. Thus, there is significant value in realizing the potential to increase student learning by tapping into the affective domain. Students may experience affective hurdles to learning that can neither be recognized nor solved when using a purely cognitive approach.

Table 1.2: A detailed presentation of the five stages of the affective domain:

Descriptors of the Major Categories in the Affective Domain	Illustrative Verbs:
Receiving phenomena: Awareness, willingness to hear, selected attention.	Examples: Listen to others with respect. Listen for and remember the name of newly introduced people. Keywords: asks, chooses, describes, follows, gives, holds, identifies, locates, names, points to, selects, sits, erects, replies, uses.
Responding to phenomena: Active participation on the part of the learners. Attends and reacts to a particular phenomenon. Learning outcomes may emphasize compliance in responding, willingness to respond, or satisfaction in responding (motivation).	Examples: Participates in class discussions. Gives a presentation. Questions new ideals, concepts, models, etc. in order to fully understand them. Know the safety rules and practices them. Keywords: answers, assists, aids, complies, conforms, discusses, greets, helps, labels, performs, practices, presents, reads, recites, reports, selects, tells, writes.
Valuing: The worth or value a person attaches to a particular object, phenomenon, or behavior. This ranges from simple acceptance to the more complex state of commitment. Valuing is based on the internalization of a set of specified values, while clues to	Examples: Demonstrates belief in the democratic process. Is sensitive towards individual and cultural differences (value diversity). Shows the ability to solve problems. Proposes a plan to social improvement and follows through with

these values are expressed in the learner's overt behavior and are often identifiable	commitment. Informs management on matters that one feels strongly about. Keywords: completes, demonstrates, differentiates, explains, follows, forms, initiates, invites, joins, justifies, proposes, reads, reports, selects, shares, studies, works.
Organization: Organizes values into priorities by contrasting different values, resolving conflicts between them, and creating an unique value system. The emphasis is on comparing, relating, and synthesizing values.	Examples: Recognizes the need for balance between freedom and responsible behavior. Accepts responsibility for one's behavior. Explains the role of systematic planning in solving problems. Accepts professional ethical standards. Creates a life plan in harmony with abilities, interests, and beliefs. Prioritizes time effectively to meet the needs of the organization, family, and self. Keywords: adheres, alters, arranges, combines, compares, completes, defends, explains, formulates, generalizes, identifies, integrates, modifies, orders, organizes, prepares, relates, synthesizes.
Internalizing values (characterization): Has a value system that controls their behavior. The behavior is pervasive, consistent, predictable, and most	Examples: Shows self-reliance when working independently. Cooperates in group activities (displays teamwork). Uses an objective approach in problem

importantly, characteristic of the learner. Instructional objectives are concerned with the student's general patterns of adjustment (personal, social, emotional).	solving. Displays a professional commitment to ethical practice on a daily basis. Revises judgments and changes behavior in light of new evidence. Values people for what they are, not how they look. Keywords: acts, discriminates, displays, influences, listens, modifies, performs, practices, proposes, qualifies, questions, revises, serves, solves, verifies.
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1.4: The Psychomotor Domain

The psychomotor domain includes physical movement, coordination, and use of the motor-skill areas. Development of these skills requires practice and is measured in terms of speed, precision, distance, procedures, or techniques in execution. Psychomotor skills range from manual tasks, such as performing a titration, to more complex tasks, such as operating a GC-MS spectrometer.

1.4.1: Rationale for Cues

A teacher's overall instructional effectiveness depends heavily on how that teacher uses instructional cues. A cue consists of a word, phrase, or sentence that describes a particular aspect of a concept or skill. While cues most often focus on motor skill development in physical education, they may also target fitness, strategy, character development, or any other aspect of lessons teachers deem appropriate. Cues enhance learning by improving student attention, comprehension, and retention.

1.4.2: Attention

Cues enhance the attention or focus of learners by restricting what they need to think about. Since learner's capacity for attention is limited, it is important to deplete the capacity with relevant, rather than irrelevant (or perhaps, less relevant) stimuli. Consequently, cues play an important role in directing student's attention toward the most critical information, and away from less critical information.

1.4.3: Comprehension

In addition to helping students attend to ideas in a lesson, cues help students comprehend ideas. Naturally, students understand concepts better when teachers communicate them clearly and developmentally. Clarity depends, in part, on using the same terms consistently. Developmental cues result from focusing on (a) process elements before product elements, and (b) basic process elements before advanced process elements. Focusing too much on product elements, such as distance and accuracy, directs learner focus away from understanding how to produce a given movement. Although emphasizing advanced process elements retains proper focus, ideas are too difficult to grasp if basic elements are not understood first.

1.4.4: Retention

While understanding ideas is important, it would serve little purpose if students did not retain the information for future application. Retention is especially enhanced when cues connect new ideas to previously learned ideas in some way (Magill, 1993). In addition, retention is related to attention and comprehension chronologically: To the degree that learners (a) attend to cues, and (b) comprehend cues, they are more likely to (c) retain the information for future use. Conversely, if students are not attentive, and do not understand cues, they are ill-prepared to apply them in the future. Certainly, one of the aims of education is to foster desire and ability for students to learn independently.

1.4.5: Examples of Cues

i. Acronyms

Acronyms help students associate the first letter of a word with a key instructional idea.

Richard **o**f **Y**ork **g**one **b**attle **i**n **v**ain

R, O, Y, G, B, I, V are the first letters representing the colors of the rainbow i.e. red, orange, yellow, green, blue, indigo and violet.

The association is even stronger when an acronym is an acrostic, i.e. when it spells a word. The more familiar acronyms become, the more associations can be retained while cueing students with the acronym only. In other words, students may initially learn to shoot a basketball with the cues, "**B**end knees, **E**lbow out, **E**yes on target, **F**ollow through"; then later the cue can be trimmed back to, "**B**end, **E**lbow, **E**yes, **F**ollow", and finally, "**BEEF**."

ii. Alliterations

Alliteration is a term that describes a literary stylistic device. It occurs when a series of words in a row have the same first consonant sound.

Finesse over Force (In mechanical workshop)

Proper Preparation Prevents Poor Performance (Continuous reading)

Proper Preparation Provides Predictable Performance (Continuous reading)

Talk to Teammates (promoting teamwork)

iii. Similes

A simile is a figure of speech that compares two different things in an interesting way e.g:

Swimming:

Breast stroke kick: Kick like a frog.

Breast stroke arms: Make a heart and cut it in half. Breast stroke arms: Scrape out pail of ice cream, eat it, and go back for more.

Butterfly kick: Kick like a dolphin.

Front crawl arms: Pull like a backward 'S'.

Sidestroke arms: Pick an apple, take a bite, put it in the other hand, and then in the basket.

Volleyball:

Look through the window as you position your hands for setting.

Sit down in a chair as you take off for blocking or spiking.

iv. Slogans

A slogan is a figure of speech in which two unlike things are explicitly compared e.g:

Failing to prepare is like preparing to fail.

No "I" in TEAM.

Nobody ever gives their best and regrets it.

Sports reveal character and build character.

Take the 't' off of can't and you can.

Yes means yes and no means no.

You miss 100% of the shots you don't take

v. Rhymes

A rhyme is the repetition of a pattern of sounds in poetry created by the alternation of long and short sounds and stressed and unstressed syllables. There are several different type of units of rhythm in poetry.

- End Rhymes. Rhyming of the final words of lines in a poem.
- Internal Rhymes. Rhyming of two words within the same line of poetry.
- Slant Rhymes (sometimes called imperfect, partial, near, oblique, off e.t.c.)
- Rich Rhymes
- Eye Rhymes
- Identical Rhymes

1.4.6: Categories of the psychomotor domain

There are seven levels (categories) of the psychomotor domain. The table below shows the levels from the simplest perception (imitation) to origination (naturalization).

Table 1:3: Categories of the psychomotor domain

Category	Example and Key words(Verbs)
Perception (awareness): The ability to use sensory cues to guide motor activity.	Examples: Detects non-verbal communication cues (imitation). Adjusts the height of the forks on a forklift by comparing where the forks are in relation to the pallet. Key Words: chooses, describes, detects, differentiates, distinguishes, identifies, isolates, relates, selects.
Set: Readiness to act. It includes mental, physical, and emotional sets. These three sets are	Examples: Knows and acts upon a sequence of steps in a manufacturing process.

dispositions that predetermine a person's response to different situations.	Key Words: begins, displays, explains, moves, proceeds, and reacts, shows, states, volunteers.
Guided Response: The early stages in learning a complex skill that includes imitation and trial and error. Adequacy of performance is achieved by practicing.	Examples: Performs a mathematical equation as demonstrated. Follows instructions to build a model. Responds hand-signals of instructor while learning to operate a forklift. Key Words: copies, traces, follows, react, reproduce, responds
Mechanism (basic proficiency): This is the intermediate stage in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.	Examples: Use a personal computer. Repair a leaking faucet. Drive a car. Key Words: assembles, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, and organizes, sketches.
Complex Overt Response (Expert): The skillful performance of motor acts that involve complex movement patterns. Proficiency is indicated by a quick, accurate, and highly coordinated performance, requiring a minimum of energy. This category includes performing without hesitation, and automatic performance.	Examples: Maneuvers a car into a tight parallel parking spot. Operates a computer quickly and accurately. Displays competence while playing the piano. Key Words: assembles, builds, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, sketches. NOTE: The Key Words are the same as Mechanism, but will have adverbs or adjectives that indicate

	that the performance is quicker, better, more accurate, etc.
Adaptation: Skills are well developed and the individual can modify movement patterns to fit special requirements.	Examples: Responds effectively to unexpected experiences. Modifies instruction to meet the needs of the learners. Perform a task with a machine that it was not originally intended to do (machine is not damaged and there is no danger in performing the new task). Key Words: adapts, alters, changes, rearranges, reorganizes, revises, varies
Origination: Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skill	Examples: Constructs a new theory. Develops a new and comprehensive training programming. Creates a new gymnastic routine. Key Words: arranges, builds, combines, composes, constructs, creates, designs, initiate, makes, originates.

CHAPTER TWO

2.0: Setting and moderating examinations

Departments must ensure that they have effective procedures for setting and moderating examination papers. This requirement could be met in the following ways:

- a. The registrar academic affairs through the Deputy Vice Chancellor will set achievable deadlines at the beginning of the semester to ensure that members of academic and administrative staff have enough time to thoroughly carry out their duties and responsibilities.
- b. The head of the department must ensure that the paper setters are fully aware of how they should set their exam papers, by when, and in what format they should be submitted to the relevant administrator. For example by:
 - i. providing the paper setter with clear and timely information
 - ii. encouraging all new and/or probationary staff to discuss their paper with a more experienced colleague before submission
 - iii. requesting that the paper setter provides a marking scheme; answers to numerical questions, indicative essay answers, and an outline of how learning objectives have been examined, in addition to the exam paper
- c. The Head of department must ensure that arrangements are in place for the moderation and scrutiny of exam papers, to consider the consistency of standards, presentation and accuracy with which an exam paper has been set. The Head of department shall constitute internal examination moderation committees.

d. The committee will:

- ii. Check on whether the setting is consistent with the Senate approved standards of examination setting.
- iii. Check for overlap between questions both within an exam paper, between exam papers for different modules, and between exam papers and coursework
- iv. Considering whether:
 - a) the questions assess the stated learning outcomes
 - b) the questions are properly structured and clearly expressed
 - c) the questions are clear and not ambiguous
 - d) the paper covers all the cognitive levels of learning
 - e) questions are appropriate in length and difficulty, and given the marks allocated to them
 - f) the marks/questions add up correctly
 - g) the correct statement for the year appears on the rubric

2.1: Classification of units

All units that are offered at the university can be classified into one of the following categories:

- 1. Introductory units
- 2. General units
- 3. Course oriented units
- 4. Advanced units

2.2: Basis of classification

Taught units are classified based on the objective of having the course in the curriculum. In the proceeding sections, the objectives of each unit classification will be outlined and the recommended examination format defined.

2.2.1: Introductory units

These are units whose objective is to give basic knowledge and skills to the learners in preparation for units which are programme specific. Examples of such units are Calculus I, Organic Chemistry I, Technical drawing I, Introduction to Computer Programming etc. Based on that objective, the course content will concentrate on the lower cognitive levels with some few concepts on the higher cognitive levels. To achieve that objective, the examination structure should follow the table below. The categories are the cognitive levels.

Category	1	2	3	4	5	6	TOTAL
% Questions	15	25	40	10	5	5	100
Marks	10.5	17.5	28	7	3.5	3.5	70

1. Memory 2. Understanding 3. Application 4. Analysis
5. Evaluation 6. Creativity

2.2.2: Course specific units

These are Units whose objective is to give the Learner knowledge and skills in the area of study e.g. Mechanical Engineering, Analytical Chemistry, and Financial Engineering. The course content should be skewed towards the higher cognitive levels. The Examination structure should have the following format:

Category	1	2	3	4	5	6	TOTAL
% Questions	10	10	40	20	10	10	100
Marks	7	7	28	14	7	7	70

1. Memory 2. Understanding 3. Application 4. Analysis
5. Evaluation 6. Creativity

2.2.3: Advanced course units

These are units whose content is skewed towards the higher cognitive levels. They will cover higher undergraduate, graduate and postgraduate courses. The learners are expected to:

- i. Specialize in the areas of their interest
- ii. Be competent in problem identification and solving
- iii. Be creative

Examples of such courses are Advanced Chemical Thermodynamics, Quantum Chemistry, Calculus III, Advanced Calculus, Advanced Quantum Physics etc. The examination structure is expected to follow the following format:

Category	1	2	3	4	5	6	TOTAL
% Questions	5	10	30	25	20	10	100
Marks	3.5	7	21	17.5	14	7	70

1. Memory 2. Understanding 3. Application 4. Analysis
5.Evaluation 6. Creativity

2.2.4: General units

These are units intended to give the learner general knowledge in all spheres of living. Examples of such areas include Business Management, Financial management, Financial Investment, Social Sciences, Leadership, Co- operate management, data analysis, Research units etc. These are units that will assist the learner to integrate all fields of knowledge for a common intended purpose. Learners have uncovered a hidden passion for a field of study because of taking such courses. Examples of such units include Communication skills, Development studies, Entrepreneurial skills etc. Such units should have the course content covering all the cognitive levels but the examination structure should concentrate more on the middle cognitive levels.

Category	1	2	3	4	5	6	TOTAL
% Questions	10	10	35	25	10	10	100
Marks	7	7	24.5	17.5	7	7	70

1. Memory 2. Understanding 3. Application 4. Analysis
5. Evaluation 6. Creativity

2.2.5: Examining the Psychomotor Domain

Some departments might find it necessary to examine the skills acquired by the Learners. Appropriate projects and practicals will be given which will be examined as per the Senate approved mode of examination

2.3: Moderation of examinations

Moderation is the process of eliminating or lessening extremes. It is used to ensure normality on the examinations being conducted. It also ensures that the expected learning objectives are attained.

2.3.1: Internal Moderation of Draft Examination Papers

Internal moderation of examinations will ensure that:

- i. The syllabus is fully covered
- ii. The examination questions cover all the cognitive categories of hierarchical learning to the expected level.
- iii. The questions are not ambiguous
- iv. The model answers are clear and correct
- v. The marks distribution is very clear and as per the approved senate document for each cognitive level and unit type.
- vi. The examination questions have not been lifted from the internet or any previously set examinations.
- vii. The questions set correctly merge the model answers as presented in the marking scheme.

- viii. The total marks are correctly merged in the question paper and marking scheme.

The HOD as per Procedure PM-260-8 Procedure for Internal/ External Moderation of Examinations, will ensure that clauses 7.1-7.3 are adhered to and parts i-iv above are effected. The moderators will have to fill the internal examination moderation form. At the end of the exercise, the HOD will compile a report which will be sent to the DVC AA.

2.3.2: External Moderation of Examinations

The University through the Office of the deputy Vice Chancellor Academic Affairs, will appoint from any other Senate approved University, an external Examiner competent in the area of the Academic programme. The appointee must be of at least Senior Lecturer grade. In accordance with PM-260-8, the external examiner will moderate the draft question papers and marked Examination scripts.

2.3.2:1 External Moderation of Draft Examination Papers

The external examiner will confirm that, for each examination paper presented:

- i. The syllabus is fully covered
- ii. The examination questions cover all the cognitive categories of hierarchical learning to the expected level.
- iii. The questions are not ambiguous
- iv. The model answers are clear and correct
- v. The marks distribution is very clear and as per the approved senate document for each cognitive level and unit type.
- vi. The examination questions have not been lifted from the internet or any previously set examinations.
- vii. The questions set correctly merge the model answers as presented in the marking scheme.
- viii. The total marks are correctly merged in the question paper and marking scheme.

At the end of the exercise, the external examiner will write a report which will address i-v above and send it to the deputy Vice Chancellor Academic Affairs.

2.3.2.2: External Moderation of Examination Scripts

The external examiner shall:

- a) For each examination scripts per each unit presented:
 - i. Systematically sample 20% of the marked scripts in cases where the scripts are above 125
 - ii. Or systematically sample 25 scripts in cases where the scripts are $25 \leq 100$
 - iii. Or sample all the scripts in cases where the scripts are below 25
- b) Write a detailed report to the Vice Chancellor which must capture the following for each of the units examined:
 - i. The general grade distribution,
 - ii. The mean mark,
 - iii. the average mean for the sample scripts,
 - iv. the standard deviation for the sample scripts,
 - v. the average mark distribution per category of cognitive domain for the sample scripts,
 - vi. a recommendation based on the course purpose based on the sample, that is, where should the examiners put more emphasis based on the cognitive levels,
 - vii. and the general performance in all the units moderated

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CHAPTER THREE

3.0: Curriculum Development and Review

The term curriculum refers to the lessons and academic content taught in a school or in a specific course or program. In dictionaries, curriculum is often defined as the courses offered by a school, but it is rarely used in such a general sense in schools. Depending on how broadly educators define or employ the term, curriculum typically refers to the knowledge and skills students are expected to learn, which includes the learning standards or learning objectives they are expected to meet; the units and lessons that teachers teach; the assignments and projects given to students; the books, materials, videos, presentations, and readings used in a course; and the tests, assessments, and other methods used to evaluate student learning.

3.1: Reasons for Curriculum Design or Review

Curriculum design or review must arise in response to a need. The curriculum so designed must therefore be a solution to an identified need. If this is the case, the graduates produced will be relevant to the Job market and hence will easily be absorbed or be self-employed. Design/review of any curriculum must:

- i. be driven by the stakeholders demand
- ii. arise from market needs assessment

There are a variety of stakeholders to JKUAT based on the charter that was granted by the Commission for University Education. These include but not limited to Industries, Farmers, Research Institutions, Financial Institutions, Health Institutions, Academics, and Human Resource Institutions. With this basic requirement in place, PM-2-58-4, Procedure for development & review of academic programmes will apply.

3.2: Commission for University Education (CUE) Standards for Curriculum Development/Review

The Commission for University has given the standard requirements for curriculum development/review. Some guidelines are here outlined:

3.3: The Nature of an Academic Programme

An academic programme shall facilitate a balanced learning process, ensuring that the Students are able to acquire such cognitive, affective and psychomotor skills as are consistent with educational goals and aspirations of Kenyans. The design of an academic programme shall ensure that the programme:

1. Is relevant and contextualized;
2. Contributes to the overall national human resource development and requirements and higher
3. Education reforms;
4. Is broad-based, diversified and integrated; and
5. Is as much as possible practical-orientation.

Each level of academic programme shall be differentiated by specific attributes. Higher levels of academic programmes shall require more complex attributes according to Bloom's Taxonomy.

3.3.1: Bachelor's degree programmes

A bachelor's academic programme is considered to be a foundational educational programme. The programme shall:

- i. Provide a broad knowledge base within a discipline involving critical and analytical understanding of the major theories, principles and concepts in the discipline;
- ii. Provide the learner with a comprehensive range of cognitive and analytical skills and their application to various situations;

- iii. Entail demonstration of adequate problem solving skills; and
- iv. Enhance society consciousness and contributions to the general development of the society.

3.3.1.1: Instructional Hours

A bachelors' academic programme shall carry a minimum number of total instructional

Hours depending on the field of study:

- i. Applied Sciences - 2240
- ii. Arts and Humanities - 1680
- iii. Medical and Allied Sciences - 3960
- iv. Pure and Natural Sciences - 1785
- v. Social Sciences - 1680

3.3.2: Postgraduate programmes

A postgraduate programme shall be offered to individuals who have completed a bachelor level degree. The levels of postgraduate programmes will comprise of:

- i. Postgraduate Certificate programme
- ii. Postgraduate Diploma programme
- iii. Master's degree programme
- iv. Doctoral degree programme

The structure of a postgraduate degree programme shall consist of various components that include taught/coursework, research and practicum. Each component of the programme shall be assessed separately.

3.3.2.1: Duration of a postgraduate programme

A postgraduate diploma/certificate programme shall carry a minimum of 360 Instructional hours and shall extend over a period of not less than one academic year.

3.3.3: Master's degree programmes

A master's programme shall be designed to provide additional education or training in the student's specialized branch of knowledge. It shall demonstrate mastery in a specific field of study or area of professional practice. The programme shall:

- i. Offer highly specialized knowledge in a discipline or field of study;
- ii. Enhance critical awareness of knowledge issues in a discipline or field of study;
- iii. Promote specialized problem solving skills required in research and/or innovation
- iv. Demonstrate mastery of a given discipline or field of study;
- v. Entail originality and creativity in the application of knowledge, skills and practice in the field of study;
- vi. Contribute to advanced knowledge and skills in the field of study; and;
- vii. Prepare the individual to take a significant responsibility in society.

3.3.3.1: Duration of master's programme

A master's degree programme shall consist of a minimum of 630 instructional hours;

- i. The structure of a master's degree programme shall demonstrate the weighting of courses in relation to coursework and thesis/dissertation;

- ii. Research shall form an integral part of a master's degree programme and shall constitute not less than one third ($\frac{1}{3}$) of the entire programme structure;
- iii. The length of the thesis/dissertation of a master's degree programme shall normally be at least 20,000 words.

3.3.4: Doctoral degree programmes

A doctoral programme shall be designed to train research scholars and, in many cases, future university staff members. The programme shall:

- i. Enhance knowledge at the most advanced frontier of the discipline or field of study;
- ii. Provide the learner with the most advanced and specialized skills and techniques required to:
 - a) Solve critical problems in research and/or innovation; and
 - b) Extend and redefine existing knowledge.
- iii. Entail demonstration of substantial authority, innovation, autonomy, scholarly and professional integrity and sustained commitment to the development of new ideas or processes at the fore-front of a discipline or field of study.

3.3.4.1: Duration of the Doctoral Degree Programme

A doctorate degree programme shall normally extend for at least three (3) academic years.

3.3.5: Types of Doctoral Degree Programmes

Two kinds of doctoral level programmes exist at the same classification level, notably; Doctor of Philosophy degrees programmes and professional doctoral degrees.

- i. Doctor of Philosophy degree programme shall focus on producing scientist practitioners who are mainly engaged in generating knowledge through research and teaching;
- ii. Professional doctoral degree programmes are normally practice-oriented and focus on producing practitioner scholars who are engaged mainly in consuming knowledge, practice and active research.

The structure of a doctoral degree programme shall demonstrate the weighting of courses in relation to coursework and thesis/dissertation

Research shall form an integral part of a doctoral programme and shall constitute not less than two thirds ($\frac{2}{3}$) of the entire programme structure;

The length of the thesis/dissertation of a doctoral programme shall normally be at least 50,000 words;

Before the award of a doctoral degree, a candidate shall normally show proof of acceptance for publication of at least two (2) papers in refereed journals.

3.3.6: Executive degrees

Executive degrees shall:

- a) Only be offered at master's level to applicants holding management positions in industry;
- b) Be a terminal degree that does not qualify the holder to teach at the university or gain entry into a doctoral programme;
- c) Be open only to candidates who meet the minimum entry requirements for a master's degree;
- d) All the requirements for a master's degree shall apply save for the thesis/dissertation which will be a project.

3.3.7: General Format for JKUAT Academic Programme

To ensure quality in all academic programmes, the Commission for University Education has guidelines of the format of the curriculum. JKUAT has adopted this format in all its programmes. Below is an outline of all the essential components:

1. Academic regulations for the programmes
 - 1.1 Admission requirements for the programme.
 - 1.2 Student assessment policy.
 - a) Continuous Assessment
 - b) End-Term/Semester
 - c) Practical
 - d) Other Assessments
 - 1.3 Evaluation of Teaching
 - 1.4 Grading system — Marks, Letter grades
 - 1.5 Examination regulations including disciplinary action and mode of appeal.
 - 1.6 Moderation of examination papers and the role of internal and external examiners.
 - 1.7 Graduation requirements.
 - 1.8 Classification of degrees.
 - 1.9 Regulations for thesis/dissertation/project.
2. Evaluation of academic staff.
3. Title of the programme offered:
 - 3.1. Spell out the specific objectives of each programme offered (programme learning outcomes).
 - 3.2. List of courses offered for the programme.
 - i. Lecture hours/course units/credit hours (as the case may be), for each, indicating core courses to be taken by a student taking a given course;

- a) Elective courses (giving details of load as above),
- b) Other courses which may be taken to meet graduation requirements (giving details of load as above);
- c) Total lecture hours/course unit/credit hours required for graduation;

3.3 Course Outline (Format)

- a) Title of the course (including code and credit hours/lecture hours/course units as the case may be),
- b) Purpose of the course;
- c) Course objectives (learning outcomes).
What the student is expected to have learned at the end of the course. The objectives should include statement knowledge, skills and attitude the student would be able to acquire, and the tasks he would be able to perform or accomplish by taking the particular course,
- d) Course description (not course outline). Course content should include all possible topics to be covered under the course;
- e) Teaching methodology; e.g. lectures, tutorials, practical's, assignments or case studies etc;
- f) Instructional materials/equipment; e.g. projectors, computers, gas chromatographs, microscope etc.
- g) Course assessment; type and weighting e.g. CAT (%) and university examination (%);
- h) Provide separately at least three references in the format:
 - i. Course Textbooks
 - ii. Course Journals
 - iii. Reference Textbooks
 - iv. Reference Journals

- Provide separately at least three Reference Textbooks and journals.
 - Provide ISBN numbers for Course and Reference Textbooks.
 - Provide ISSN number for Course and Reference Journals.
- v. Appendices of the academic resources for the support of the programme comprising of the facilities; equipment and teaching materials; academic staff.

3.3.8: Note

There shall be a linkage between the various facets of an academic programme:

- a) The institutional information shall be cascaded down to the programme, while the programme's general information shall be cascaded down to the various courses of the programme;
- (b) The course title shall be reflected in the purpose of the course and the course's expected learning outcomes; and
- c) The expected learning outcomes shall be reflected in the course content, which shall be linked to the mode of delivery, instructional materials and/or equipment, assessment and reference materials.

3.3.9: Academic Programme Title

The academic programme title shall be reflective of its overall content.

Guidelines:

1. The academic programme title shall adapt universal nomenclature for similar programmes;
2. The academic programme title shall be short, concise and descriptive of its overall content;

3. The appropriateness of an academic programme title shall be determined by:
 - a) At least 70% of the content being relevant and core to the academic programme. This includes both core and specialization courses; and
 - b) For an academic programme title that include two (2) field of study, the allocations of core courses to the two (2) fields shall be equally shared.
4. Two (2) or more similar academic programmes shall be considered distinct, thus warranting unique programme titles, if more than 30% of the core content is different

3.3.10: Expected Learning Outcomes

Curriculum development is a response to the stakeholder's needs. The University through a relevant department, develops a programme, with an objective of giving a solution to the need. This involves teaching and training learners to have the required knowledge and skills. At the end of the training period, the Learners must be examined to determine whether the objective was achieved.

There are three domains in Learning. Most examinations test the cognitive domain. Out of the three domains, the affective domain is the hardest to test in the ordinary examinations that are normally given in the Universities.

The expected outcomes must be SMART i.e specific, measurable, attainable within a set time and realistic. Based on the unit classification, the expected outcome must constitute a verb based on the cognitive level of the course objective. For example, introductory units will be based on the lower cognitive levels. Suppose learners are being introduced to causes and effects of HIV/AIDS, the Learner will be expected to remember the definition of HIV/AIDS, name the virus that causes AIDS, understand how the Virus causes aids. The Learning outcomes must be based on the Lower cognitive levels. For example, some of the expected outcomes will be:

- i. The Learner should be able to **name** the Virus that causes AIDS
- ii. The Learner should be able to **draw** the Virus that causes AIDS
- iii. The Learner should be able to **list** the possible ways of getting HIV
- iv. The Learner should be able to **explain** how HIV causes AIDS
- v. The Learner should be able to **contrast** the HIV and other types of viruses.

In the above examples, the Memory and Understanding cognitive levels have been tested.

The key words for writing the expected outcomes based on the cognitive domain can be chosen from table 1.1 based on the level of testing. The psychomotor domain can also be tested and the key words for the expected outcome chosen from table 1.3. Suppose a chemistry student wants to determine the amount of $\text{CH}_3\text{CH}_2\text{OH}$ in an analyte, what form of spectrometry should he/she use given an option between the Infrared (IR) spectrometer and the Atomic Absorption Spectrometer?. This will require a cognitive distinction of the transition modes available in the given molecule requiring proper Perception (**awareness**) i.e the ability to use sensory cues to guide motor activity. The expected Learning outcome should be '*the Learner should be able to choose the right Spectrometer for analysis of chemical systems*'. This is the lowest level of the psychomotor domain. The other levels of the psychomotor domain can be tested by making the right choice of the verb.

CHAPTER FOUR

4.0: Monitoring and Evaluation

At various stages of programme development and implementation, evaluation studies are undertaken to provide direction to the programme. Monitoring is concerned largely with ensuring that inputs lead to the intended outputs. Evaluation attempts to determine the relevance, effectiveness, efficiency and effects of programme activities in the light of the objectives of the programme.

Evaluation is an essential activity, which feeds back into programme implementation. It is a learning and action oriented tool for improving activities. Without a monitoring and evaluation component, an educational programme is incomplete.

4.1 : Definitions of Evaluation and Related Concepts

Tyler (1950): Evaluation is a process of comparing performance data with clearly specified objectives.

The encyclopedia of Educational Research: Evaluation in education signifies describing the curriculum in terms of selected attributes and judging the degree of acceptability or sustainability.

Beeby (1977): Evaluation is the systematic collection and interpretation of evidence, leading as part of the process to a judgement of value with a view to action.

Evaluative data should be collected in a systematic manner using a variety of well-designed evaluation instruments and carefully interpreted by the evaluator. The evaluator must make judgement about the value of the curriculum and programme being evaluated. These definitions imply that evaluation is a management tool which aids programme staff to identify implementation problems and to address such problems so as to allow implantation to proceed as planned.

4.2: Measurement

Measurement is the systematic collection and orderly arrangement of educational information. It is an assignment of an exact and quantitative number such as 55 marks to an object or a trait. No value is normally placed on what is being measured.

4.3: Assessment

Assessment in education is a way of finding out to the extent to which the Learners have acquired the knowledge, attitudes and skills. Some attempt is made to apply a standard or criterion to the information about a learner's performance.

4.4: Research

Research is a careful, diligent and critical investigation of the phenomena. Basic research is directed towards generating new knowledge while applied research is directed towards application of knowledge.

4.5: Monitoring

Monitoring can be defined as follows:

Commonwealth Secretariat, 1993: Monitoring is collecting information at regular intervals about an on-going project or programme within the school system concerning the nature and level of performance.

UNESCO, 1986: To monitor a programme is to collect information in relation to the quantitative aspect of the programme implementation including logistics.

UNFPA, 1990: Monitoring is a continuous and periodic review and overseeing by programme management to ensure that project input deliveries, work schedules, targeted outputs and other required action proceed according to plan.

Monitoring enables programme directors and managers to receive frequent and up-to-date answers to the following questions:

- i. Is the curriculum/programme being implemented as planned, on schedule and within budget?
- ii. Are the curriculum/programme activities leading to expected results?
- iii. Is there anything happening that can lead management to revise/modify the original plans?
- iv. What factors are hindering the implementation of the curriculum/programme?

Monitoring is concerned with finding out the extent to which curriculum/programme activities are being implemented compared with what was planned. Evaluation seeks to explain and measure the level of efficiency.

4.6: Important Evaluation Concerns

There are aspects one has to ask questions about when assessing the performance of a programme. These are:

- i. *Progress*: this is concerned with continual development towards achieving of the objectives of the programme,
- ii. *Adequacy*: are the programme objectives and inputs enough for the purpose?
- iii. *Relevance*: Are the programme objectives and activities responding to the needs of the stakeholders?
- iv. *Effectiveness*: This refers to the extent to which a programme is producing the desired results. Effectiveness measures the degree of attainment of the pre-determined objectives of the programme.
- v. *Efficiency*: this an expression of the extent to which the methods used by the programme or activity are the best in terms of their cost, resources used, time required and appropriateness of the task. It examines whether there are adequate justification for the

resources used and identifies alternative strategies to achieve better results with the same inputs.

- vi. *Impact*: measurement of impact is concerned with determining the overall effect of a programme in terms of the socio-economic and other aspects of the community.
- vii. *Cost-effectiveness analysis*: Refers to the evaluation of alternatives according to both their costs and their effects with regard to producing an outcome or set of outcomes.
- viii. *Sustainability*: this examines the extent to which the programme strategies and activities are likely to continue to be implemented after the withdrawal of the external assistance.
- ix. *Unintended Outcomes*: these are unforeseen negative or positive effects of a project or programme.
- x. *Alternative strategies*: alternative strategies to solving the identified needs or problems are analysed and recommended to the next phase of programme review if the original strategy is found inappropriate.
- xi. *Cost Benefits*: cost benefit analysis compares the financial costs of a programme to the financial benefits of that programme. It is normally conducted on more than one programme.
- xii. *Validity of Design*: this assesses the extent to which the programme design:
 - a. Sets out clear immediate objectives and indicators of the objectives
 - b. Concentrates on the identified problems and needs and clearly spells out the strategy to be followed for solving the problems and meeting the identified needs.

- c. Describes the main inputs , outputs and activities needed to achieve the objectives
- d. States the means of verification of achievements of objectives and valid assumptions about the major external factors affecting the project.

4.7: Monitoring and Evaluation Indicators

An educational indicator is a statistic about the educational system that reveals information about its performance. Indicators are objective and specific measures of changes or results expected from a programme, project or educational activity. Indicators must be valid and reliable. Validity is the to which an indicator measures what is meant to be measured. Reliability means that the conclusions based on the use of the indicators are the same for anybody who uses it.

4.7.1: Output indicators

Not all aspects of a project, or programme are reliable for the construction of indicator. The following is a list of some possible output indicators:

- i. Gross enrollment ratio
- ii. Completion rate
- iii. Progression rate
- iv. Employment rate
- v. Attendance rates

4.7.2: Input Indicators

- i. Staff- student ratio
- ii. Number of equipment

- iii. Number of lecture rooms

Roles of Evaluation in Education

Educational Evaluation is meant to provide useful data to decision makers to enable them make decisions and judgements about the education activity evaluated. It plays many roles in:

- i. Identifying educational needs
- ii. Monitoring programme implementation
- iii. Updating educational materials, programmes and procedures
- iv. Justification of reviewing or discontinuation of a programme.

Identifying educational needs: before a new programme is developed, it is vital to identify the target learners.

Monitoring programme implementation: Monitoring is conducted to ensure that the input deliveries, work schedules and output valuables proceed according to plan.

Updating educational materials, programmes and procedures: it is important to know whether the stakeholder's needs are met. Determine the strengths of the activities on the learners and review them if necessary.

Determining whether the programme is worth: Evaluation provides information about the programme worth. Determine whether the programme objectives are attained.

4.8: Levels of Evaluation

Educational evaluation in a university system is carried out at the following levels:

a. Classroom evaluation

This is conducted by lecturers to obtain feedback on the progress being made by individual students. This is purposely carried out to improve instruction. Some of the assessment methods used are:

- i. Written tests
- ii. Oral questions
- iii. Assignments
- iv. Practicals

b. End semester Evaluation

At the end of every semester, the university conducts end semester examinations. These are set and moderated by both internal and external examiners. The examiners are specialized in the areas of examination.

4.9: Monitoring of effective teaching

Monitoring is an essential component of curriculum and programme implementation. It should not be perceived a witch hunting or fault finding activity. The following will be employed in monitoring of effective teaching:

- i. Student class attendance registers
- ii. Unit quality assurance forms
- iii. Appointment of class representatives
- iv. Unit timetabling

CHAPTER FIVE

5.0: Curriculum evaluation

Curriculum is defined as all that is planned to enable learners to acquire desirable knowledge, skills and attitudes. Once a curriculum is developed, continuous evaluation of the curriculum materials must be undertaken. The curriculum developers must also evaluate teaching effectiveness. Data obtained from the evaluation by the developers will enable them to determine the extent to which learning is taking place.

5.1: Definition of curriculum evaluation

Curriculum evaluation is the process of collecting, analyzing and interpreting data for enabling decision making at various stages of curriculum development and implementation process. Evaluation of curriculum materials such as syllabi, textbooks, laboratory equipment goes on continuously. Curriculum developers must systematically gather evidence during all stages of the curriculum development and implementation processes for the purpose of improving the curriculum.

5.2: Functions of curriculum evaluation

a) Identifying Educational needs

Curriculum evaluation crystalizes the educational needs. Before development of a new curriculum or review of an existing curriculum, it is necessary to carry out a needs assessment to identify the appropriate educational objectives. A University curriculum must be a reflection of the felt needs of the community, region or country.

b) Curriculum Revision

Curriculum evaluation justifies curriculum review. The process is known as formative evaluation whose aim is to improve on the existing curriculum. One of the major focus is on evaluation of the curriculum

content and objectives. One major objective of formative evaluation is to identify major difficulties experienced by implementers of the curriculum and the causatives. Summative evaluation involves evaluation of a completed programme.

c) Checking if the objectives have been achieved

Curriculum evaluation helps to assess if the intended curriculum objectives have been achieved. It is a means of determining whether the curriculum has enabled the learners to acquire the anticipated knowledge, skills and attitudes.

5.3: Steps in Curriculum Evaluation Process

- i. Specification of the curriculum objectives or curriculum materials to be evaluated.
- ii. Specification of the evaluation objectives
- iii. Designing of the evaluation instruments or adapting the existing instruments.
- iv. Collection of data
- v. Analysis of data
- vi. Forwarding information to the decision makers

a. Specification of the curriculum objectives or curriculum materials to be evaluated

The evaluator must specify the objectives and the subject matter of the curriculum together with the target group. For example before a text book is evaluated, it must be determined whether it is a student's book or a lecturer's guide for the unit it is intended.

b. Specification of the evaluation objectives

Identify the decision makers who will use the evaluation data the kind of decisions expected. The decision makers include the subject professionals,

the departmental board and the senate. The kind of decisions for which the evaluation data is needed include:

- i. Whether to review the syllabus or not
 - ii. Whether to approve a given textbook as a course book, reference book or not
 - iii. Whether a given course requires practicals or not
- c. Designing of the evaluation instruments or adapting the existing instruments.**

Evaluation of curriculum materials must be made with the aid of a checklist. Some kind of evaluation instrument must be used to evaluate curriculum materials. The curriculum materials to be evaluated include:

- i. Curriculum designs
- ii. Syllabi
- iii. Reading materials
- iv. Teaching facilities

These materials are evaluated in regard to aspects such as:

- Relevance to the learners needs
- Coverage of the relevant topics
- Coverage of all the learning domains
- Likely contribution to the achievement of the programme objectives
- Their sequencing

d. Collection of the evaluation Data.

A curriculum evaluation committee must be established in every department. This should be composed of specialists in the subject matter of the curriculum being evaluated. This must be headed by the chairman of the department. The curriculum materials can be evaluated using

questionnaires for lecturers and students, interviews and discussions with lecturers and students and content analysis of the materials.

e. Analysis of Data

Analysis of data is done during and after the data collection.

f. Provision of information to the Decision Makers

A report will be prepared by the evaluation committee and sent by the Chairman to the director Academic quality assurance (DAQA) who will further forward it with recommendations to the deputy vice chancellor Academic Affairs. The report must comprise strengths and weaknesses of the curriculum being evaluated. All conclusions and recommendations must be backed by evaluative data. Implementation of the recommendations will depend on several factors including the following:

- i. Quality of the evaluation
- ii. Availability of funds
- iii. The timing of the evaluation.

5.4: Role of evaluation in curriculum development

Curriculum evaluation is a continuous process and an integral part of curriculum development. A new curriculum must include a provision for systematic evaluation of curriculum materials in order to provide the curriculum specialists with evidence of the strengths and weaknesses of curriculum designs, syllabi, recommended reading materials, learning and teaching materials, physical infrastructure etc. The role of evaluation in curriculum development is as discussed in the three stages below:

i. Conceptualization

At this stage, the curriculum specialists conduct a needs assessment to determine the knowledge, skills and attitudes that students should learn.

Needs assessment will lead to the following information and decision concerning the curriculum:

- identification of the objectives and content of the new curriculum
- the resources required for the curriculum implementation.

The following must be investigated:

a) Needs of the society where the programme is implemented

It is important for the specialists to have a clear detail of what the society and the country at large will want the students to learn. This can be achieved through questionnaires, interviews, situational analysis of the existing curriculum and policy documents.

b) Needs of the students

A questionnaire is developed to collect data from potential students about the following:

- Who are they?
- What knowledge, attitudes and skills do they already have?
- What are their views on the proposed curriculum?

c) Existing curriculum

It is important to critically examine the existing curriculum to determine the extent to which it needs revision to meet the needs of the society/ stakeholders.

ii. Planning

At this stage, evaluation focuses on objectives and content of the curriculum, learning activities, learning resources, evaluation methods and time allocation. The evaluation questions to be asked include:

- ✓ Are the objectives achievable and realistic?
- ✓ Are all the topics included in the syllabus necessary?
- ✓ Is the sequence of the subject content correct?

iii. Material preparation

This stage involves preparation of instructional materials such recommended text books and journals, laboratory instruments and equipment, workshop equipment and attachment hospitals and research institutions. Instructional materials are evaluated in regard to:

- Relevance to the needs of the learners
- Suitability for the age , ability and experience of the learners
- Coverage of the relevant topics
- Coverage of the knowledge, attitude and skills domains
- Accuracy of the content
- Sequencing

iv. Curriculum Implementation

During this stage, effectiveness of the new curriculum is maintained by conducting systematic evaluation of the curriculum. This is referred to **quality control**.

5.5: Evaluation of Teaching Effectiveness

There are several reasons for evaluating teaching which include:

a. To diagnose and help teaching

The academic affairs division through the directorate of academic assurance are expected to develop tools for evaluation of lecturers in order to identify factors that may prevent them from teaching effectively. The lecturer being evaluated should be informed of his/her strengths and weaknesses. The sources of information include:

- Current and former students
- Colleagues
- Supervisors

The academic division should organize for pedagogical clinics which are mandatory. This will develop and enhance teaching skills.

b. Making decisions about individual lectures

This will include promoting the lecturer on merit and using the lecturer as a resource person for peer training

c. Determining the extent of curriculum implementation

One of the effective ways of obtaining information on curriculum implementation involves assessment of the teaching skills and competency of lectures as demonstrated by them in lecturer room situations. It is necessary to visit individual lecturers in class.

CHAPTER SIX

6.0: Programme evaluation

6.1: Meaning and importance of programme evaluation

A programme is an organized set of activities directed towards achieving specific long term objectives. Programme evaluation is the process of systematic collection, analysis and interpretation of programme related data that can be used to understand how the programme is functioning. Evaluation of educational programmes is a continuous process.

6.2: Purposes of Programme Evaluation

Programme evaluation serves many purposes including the following:

- i. To document internal programme's process
- ii. To provide systematic evidence of the programme's success or failure
- iii. To discover the effect of the programme on the receiver and propose future alterations and improvements
- iv. To document client satisfaction with the programme
- v. To conduct cost –benefit analysis of the programme
- vi. To appraise the quality of educational programmes

Programme evaluation serves both formative and summative functions

6.3: Objectives of programme Evaluation

Programme evaluation is carried out to answer questions on various aspects of the programme including:

- i. Programme objectives
- ii. Curriculum materials
- iii. Curriculum implementation
- iv. Student assessment
- v. Student's performance

- vi. Staffing
- vii. Lecturer performance
- viii. Teaching and learning recourses
- ix. Didactic methods
- x. Financing of the programme

These aspects are analyzed to provide data that can used by decision makers to improve the quality of the programme.

6.4: Levels of programme Evaluation

Programme evaluation involves five levels:

- a. Needs assessment
- b. Programme monitoring
- c. Formative evaluation
- d. Summative evaluation
- e. Impact assessment

6.5: Purposes of Programme monitoring

- i. To receive regular feedback on how the programme is working
- ii. To provide the basis for corrective measures to be undertaken through early detection of problems and performance gaps
- iii. To verify proper utilization of programme resources
- iv. to ensure that inputs are availed at the right time and transformed effectively into outputs through the proposed activities
- v. to identify the problems early
- vi. To enable interpretation of the findings of the formative, summative and impact evaluation studies.
- vii. To enable the managers to assess the extent to which the programme is being implemented
- viii. To identify unexpected results during early stages and make appropriate corrections

6.6: Formative Evaluation

Formative evaluation is carried out during the operation of an educational programme. This is intended to provide staff with data in order to improve programme operations. It is also referred to as process evaluation. The following questions are posed during the evaluation:

- i. What are the programme objectives and to what extent are they realistic and achievable?
- ii. To what extent is the programme effectively and efficiently managed?
- iii. Are the activities of the programme leading to the achievement of the objective?
- iv. What are the programme strengths and weaknesses?
- v. What changes can be effected in order to improve the programme?
- vi. What challenges are the lecturers and learners encountering?
- vii. What are the barriers to the programme implementation?

The main objective of formative evaluation is to identify the aspects of the programme where revision is needed.

6.7: Summative Evaluation

Summative evaluation is carried out when the programme has gone through its full cycle or when the programme is nearing completion. This is meant to objectively assess the effects of the programme, document programme effectiveness and provide data for programme improvement. Some of the common questions to be asked include:

- a. What are the goals and objectives of the programme?
- b. Are the goals of the programme realistic and achievable?
- c. Was the programme implemented as planned?
- d. How effective is the programme in comparison with other programmes?
- e. How costly is the programme?

Summative evaluation involves the following six stages:

1. Assessment of effectiveness- to assess the degree of achievement of objectives, targets and expected results
2. To assess relevance- are the planning elements relevant to the national education priorities?
3. Assessment of efficiency- this measures the results achieved in relation to the resources used
4. Assessment of quality- this measures the extent to which the results meet technical/professional/ stakeholders expectations/standards.
5. Assessment of acceptability- this is the extent of appreciation of the programme results by the stakeholders
6. Determination of the impact- this refers to the total, direct and indirect effects of the programme and social economic development of the target group.

APPENDICES

Appendix One: Departmental Quality Assurance Committee

Each academic department will appoint a quality assurance committee of five members chaired by the chairman of the department. The terms of reference for this committee will be:

1. To evaluate the curriculum at the end of every academic year and write a self-assessment report.
2. To evaluate the programme at the end of the full cycle and write a self-assessment report.
3. To create mechanisms of monitoring effective teaching and learning in the department.
4. To write recommendations to the senate through the Director DAQA on the programme.
5. Any other matters touching on quality assurance

Appendix Two: Curriculum Evaluation Tool

- a) Each annual curriculum evaluation (formative) report will comprise the following:
 - 1. Title of the programme e.g B.Sc Analytical Chemistry
 - 2. Objectives of the programme
 - 3. A summary on the relevance of the curriculum content towards meeting the programme objectives
 - 4. A summary on the evaluation of the curriculum materials i.e reference materials, equipments etc. towards meeting the programme objectives.
 - 5. Number of students admitted vis a vis those who reported in first year
 - 6. General performance in all levels, that is, 1st year, 2nd year e.t.c. and progression rate.
 - 7. Didactic teaching and Learning methods and whether there is room for improvement.
 - 8. Recommendations to the way forward on the status and feed forward of the programme.

- b) Programme evaluation (summative) report will be conducted at the end of the full cycle as self-assessment report shown in appendix three:

Appendix Three: Self-assessment report



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY**

DEPARTMENT OF (e.g. CHEMISTRY)

**SELF ASSESSMENT REPORT FOR (e.g. BSc. INDUSTRIAL
CHEMISTRY)**

ACADEMIC YEAR (e.g. 2016/2017)

TABLE OF CONTENTS**1. INTRODUCTION**

- 1.1** Background Information
- 1.2** Vision Statement
- 1.3** Mission Statement
- 1.4** Motto
- 1.5** Departmental Objectives

2. EXPECTED LEARNING OUTCOMES**3. PROGRAMME CONTENT****3.1 YEAR ONE**

S/NO	UNIT CODE	UNIT NAME	CONTACT HOURS

e.t.c.

**4. DIDACTIC CONCEPT/TEACHING/LEARNING
STRATEGY****5. STUDENT ASSESMENT****6. ACADEMIC STAFF QUALITY**

- 6.1** Preamble
- 6.2** Details of Staff

S/NO	NAME OF STAFF	QUALIFICATION

6.3 Explanation of Qualifications if any.

7. TECHNICAL STAFF AND ADMINISTRATIVE STAFF INVOLVED IN THE PROGRAMME

7.1 Preamble

7.2 Details of Staff

S/NO	NAME OF STAFF	QUALIFICATION

8. STUDENT'S PROFILE

8.1 Admission Criteria

8.2 Transition rate

- i. Show the status eg. for 2015/2016 number of students from 1st year to the final year
- ii. Comment on transition rate
- iii. Comment on admission vs reporting rate for first years
Show data using tables and bar charts where appropriate

8.3 Information on Alumni (Results of Tracer Study done for the programme)

9. STUDENT ADVISORY / SUPPORT SERVICES

10. FACILITIES AND INFRASTRUCURE AVAILABLE FOR THE PROGRAMME (Compare with the CUE requirements)

11. CURRICULUM DESIGN, REVIEW AND EVALUATION

12. STAFF DEVELOPMENT ACTIVITIES

13. BENCHMARKING

**14. ACHIEVEMENTS/IMPACTS OF THE PROGRAMME
AND ITS GRADUATES**

**15. REQUIREMENT AND SATISFACTION OF
STAKEHOLDERS**

16. RECOMMENDATIONS ON THE WAY FORWARD

Appendix Four: Internal draft examination moderation form

Internal draft examinations moderation will be done by all departmental academic staff of Lecturer grade and above.

The examination moderator will ensure that:

- i. The syllabus is fully covered
- ii. The examination questions cover all the cognitive categories of hierarchical learning to the expected level.
- iii. The questions are not ambiguous
- iv. The model answers are clear and correct
- v. The marks distribution is very clear and as per the approved senate document for each cognitive level and unit type.
- vi. The examination questions have not been lifted from the internet or any previously set examinations.
- vii. The questions set correctly merge the model answers as presented in the marking scheme.
- viii. The total marks are correctly merged in the question paper and marking scheme.

The format of the moderation form will be as below:

INTERNAL DRAFT EXAMINATION MODERATION FORM

1. Department.....
2. Unit Code & Name
3. Level e.g. Diploma, B.Sc., Masters.....
4. Year e.g First Year.....
5. Academic Year/ Semester e.g 3rd Year/ Sem II
6. Unit Classification e.g Introductory.....

S/NO	Cognitive level	% Mark awarded to questions	% Mark recommended by DAQA	Approved/Not Approved
1	Memory			
2	Understanding			
3	Application			
4	Analysis			
5	Evaluation			
6	Creativity			
7	Total Marks			

7. Do the questions fully cover the syllabus? Yes/NO.....
8. Do the examination questions cover all the cognitive categories of hierarchical learning to the expected level? Yes/NO.....

9. Are the questions Clear? Yes/NO.....
10. Are the model answers clear and correct? Yes/NO.....
11. Is the marks distribution clear and as per the approved senate document for each cognitive level and unit type? Yes/ No.....
12. Are all the examination questions lifted from the internet or any previously set examinations? Yes/NO.....
13. Do the questions set correctly merge the model answers as presented in the marking scheme Yes/No.....

Recommendation.....

Chairman of :

Panel.....Sign.....Date.....

COD Sign.....Date.....

Appendix Five: External Examination moderation

External Moderation of Examinations.

The University through the Office of the deputy Vice Chancellor Academic Affairs, will appoint from any other Senate approved University, an external Examiner competent in the area of the Academic programme. The appointee must be of at least Senior Lecturer grade. The external examiner will moderate both the draft question papers and marked Examination scripts.

External Moderation of Draft Examination Papers

The external examiner will confirm that, for each examination paper presented:

1. The syllabus is fully covered
2. The examination questions cover all the cognitive categories of hierarchical learning as per University approved standards.
3. The questions are not ambiguous
4. The model answers are clear and correct
5. The examination questions have not been lifted from the internet or any previously set examinations.
6. The questions set correctly merge the model answers as presented in the marking scheme.
7. The total marks are correctly merged in the question paper and marking scheme.

At the end of the exercise, the external examiner will write a report which will address the aspects in 1-6 above and send it to the deputy Vice Chancellor Academic Affairs.

External Moderation of Examination Scripts

The external examiner shall:

a) For each examination scripts per each unit presented:

i. Systematically sample 20% of the marked scripts in cases where the scripts are above 125.

ii. Or systematically sample 25 scripts in cases where the scripts are $25 \leq 100$

iii. Or sample all the scripts in cases where the scripts are below 25

b) Write a detailed report to the Vice Chancellor which must capture the following for each of the units examined:

a) The general grade distribution,

b) The mean mark,

c) the average mean for the sample scripts,

d) the standard deviation for the sample scripts,

e) the average mark distribution per category of cognitive domain for the sample scripts,

f) a recommendation based on the course purpose based on the sample i.e where should the examiners put more emphasis based on the cognitive levels,

g) and the general performance in all the units moderated

Appendix six: Internal Academic Quality Assurance Auditing

The directorate of quality assurance will undertake random auditing of the teaching departments to confirm whether they adhere to the University senate approved academic assurance standards. The auditing will cover both infrastructure and service delivery. The following audit questionnaire will be used.

1. **DEPARTMENT**.....
2. **DATE OF AUDITING**.....
3. **NAME OF THE PROGRAMME**.....
4. **AUDIT SCOPE**.....
5. **AUDIT CRITERIA**.....
6. **CURRICULUM IMPLEMENTATION**
 - i. When was the curriculum developed/ reviewed.....
 - ii. Is there any formative programme evaluation carried out.....
 - iii. Is the programme due for review.....
 - iv. As summative evaluation been carried out.....
7. **EXAMINATION SETTING AND MODERATION**
 - i. Does the department have records of internal and external of examinations.....
 - ii. Are examinations internally moderated as per the approved Senate approved University standards?.....
 - iii. Are the examinations externally moderated as per the senate approved University standards?.....

- iv. Are their recommendations by the external examiner?.....
- v. Were the external examiners' recommendations addressed?.....

8. PROGRAMME FACILITIES

- i. What is the total number of students admitted into the department?.....
- ii. What is the total number of students admitted into the programme?.....
- iii. What is the student / Lecturer ratio in the deparment for all existing programmes?.....
- iv. Is the laboratory space available as per the CUE recommendations?.....
- v. Is the office space as per the CUE recommendations?.....

Appendix Seven: Standard for setting a common examination

A common unit is a unit with the same code and content offered to various campuses/ colleges/ centers at the same period of the academic year. The unit could be taught by the same or different Lecturers. The university Policy stipulates that students undertaking such a unit will sit a common examination. The common examination shall:

- i. Be set and moderated by a panel of examiners
- ii. Be done at the same time in all the centers where it will be conducted
- iii. Be marked by the individual Lecturers teaching the common unit from each center

Setting and moderating a common examination

- i. The coordinator and chairman of the setting and moderating of the common examination shall be chairman/ Chairperson of the department offering the common unit in Jomo Kenyatta University Juja campus.
- ii. There shall be at least three panelists, chosen from the different centers offering the common unit.
- iii. The panelists will be shall be individuals who are teaching the unit at the time of offer or who have previously taught the unit.
- iv. The examination will cover the whole syllabus as per the University standards stipulated the Quality assurance handbook.
- v. The panel shall set two papers. These shall cover Ordinary examinations and special/supplementary examinations.
- vi. The panelists shall provide a detailed marking scheme which will be distributed to all centers offering the common unit.

External moderation of the common examination

- i. All draft common examinations shall be moderated by one external examiner appointed by the senate

- ii. All student scripts shall be moderated by one external examiner appointed by the senate.
- iii. The coordinator and chairman of the external moderation of the common examination shall be chairman/ Chairperson of the department offering the common unit in Jomo Kenyatta University Juja campus.
- iv. All directors/ chairmen from the different centers shall ensure that the external examiner receives the scripts as per the schedule.

Supplementary for common examinations

- i. There shall be one common Special/supplementary examination for all common examinations
- ii. The common special / supplementary examinations shall be conducted as per the senate approved dates
- iii. Any student who does not sit for ordinary or special / supplementary examinations as per the senate approved dates will sit for the same in senate approved subsequent dates for subsequent academic years.

Common examination regulations

University Common examination regulations will always apply.